

Architectural Specification – FinTrend Analytic Platform

This document describes the **High-Level Design (HLD)**, **Low-Level Design (LLD)**, and **System Flow Diagrams** for the FinTrend Analytic Platform, an enterprise-grade mutual fund portfolio analytics and compliance manager app.

1. High-Level Design (HLD)

The FinTrend Analytic Platform is architected as a React-based single-page application (SPA). It uses a clean, component-driven client architecture to maximize performance, visual responsiveness, and secure local data state isolation.

![High-Level Design (HLD) Flow Diagram](/Users/maa/.gemini/antigravity/brain/9acfab5c-42be-48ea-b1d7-73a3c666e67e/hld_diagram_technical_1782580548990.jpg)

HLD Design Strategy

1. **Presentation Layer**: Built on React 19 and custom HSL variables in `index.css`. The layout leverages a responsive sidebar grid that collapsed to uppercase abbreviations. All layout items enforce a sharp square border theme (`border-radius: 0px`).
2. **Controller/Routing Layer**: Handled by `react-router-dom` to coordinate routes across dashboards.
3. **Data State & Persistence**: Uses standard React hooks for state and `localStorage` to simulate an online relational database of managers and session values, making testing client-isolated and lightweight.

2. Low-Level Design (LLD)

The LLD breaks down individual file structures, core responsibilities, and API-like function mappings.

![Low-Level Design (LLD) Class Diagram](/Users/maa/.gemini/antigravity/brain/9acfab5c-42be-48ea-b1d7-73a3c666e67e/lld_diagram_simple_1782580355317.jpg)

Component Details

- * **[App.jsx](file:///Users/maa/Desktop/analysis/TOTALAnalytics/src/App.jsx)**: Root container. Initializes states. Coordinates the loading splash and authenticates routes.
- * **[Auth.jsx](file:///Users/maa/Desktop/analysis/TOTALAnalytics/src/components/Auth.jsx)**: Implements validation login checks against a pre-seeded local storage dataset. Saves new registration models (`Manager Name`, `Phone`, `User ID`, `Age`, `Password`, `Email`, `State`, `City`).
- * **[PerformanceBenchmarking.jsx](file:///Users/maa/Desktop/analysis/TOTALAnalytics/src/components/PerformanceBenchmarking.jsx)**: Compiles portfolio returns, alphas, and ratios into Chart.js elements. Includes absolute endpoint overlays for lines rendering and conditional insight bullet grids.
- * **[index.css](file:///Users/maa/Desktop/analysis/TOTALAnalytics/src/index.css)**: Implements theme variables, custom card structures, and a global reset `* { border-radius: 0px !important; }`.

3. End-to-End System Flow

The diagram below details the sequence from initial user arrival to interactive dashboards and profile actions.

![E2E System Sequence Flow Diagram](/Users/maa/.gemini/antigravity/brain/9acfab5

c-42be-48ea-b1d7-73a3c666e67e/sequence_diagram_simple_1782580373674.jpg)

4. Sequential API Gateway Flow

The diagram below details the sequential API transactions and client-to-storage data flow exchanges for logins, queries, chat requests, and spreadsheet compilation:

![Sequential API Flow Diagram](/Users/maa/.gemini/antigravity/brain/9acfab5c-42be-48ea-b1d7-73a3c666e67e/api_flow_diagram_light_1782579995028.jpg)

5. UI Visual Mockups (Light Mode / White Backgrounds)

````carousel

![FinTrend Dashboard Mockup](/Users/maa/.gemini/antigravity/brain/9acfab5c-42be-48ea-b1d7-73a3c666e67e/fintrend\_dashboard\_mockup\_1782579268908.jpg)

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![FinTrend Login Mockup](/Users/maa/.gemini/antigravity/brain/9acfab5c-42be-48ea-b1d7-73a3c666e67e/fintrend\_login\_mockup\_1782579288012.jpg)

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